Cloud Computing

Lecture

Introduction to Cloud Computing

Our story…

Our Data Now…

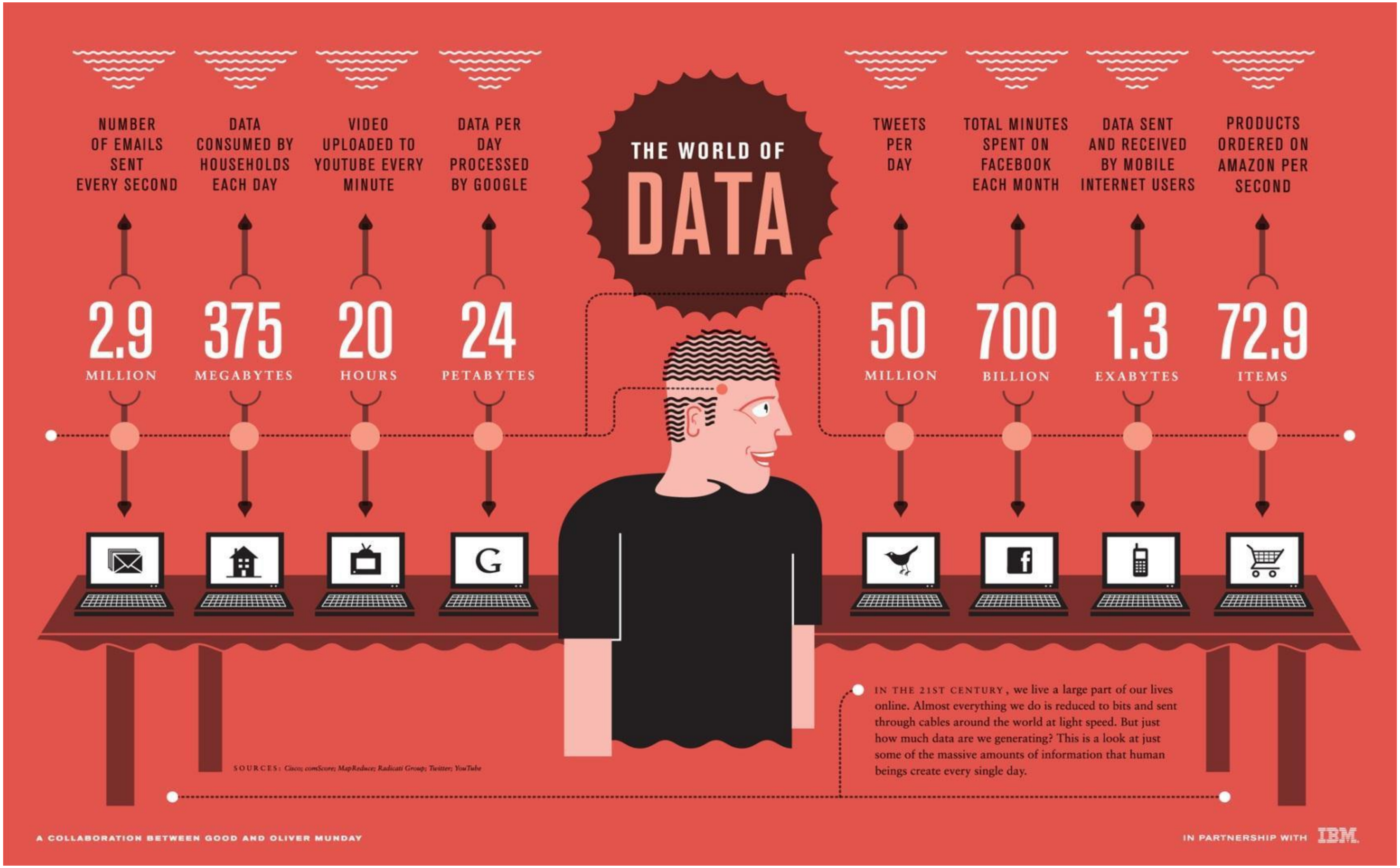
Documents and Media

Personal Data

Emails, Calendars, Contacts, Location Information, etc…

We Live in a World of Data…

The World of Data



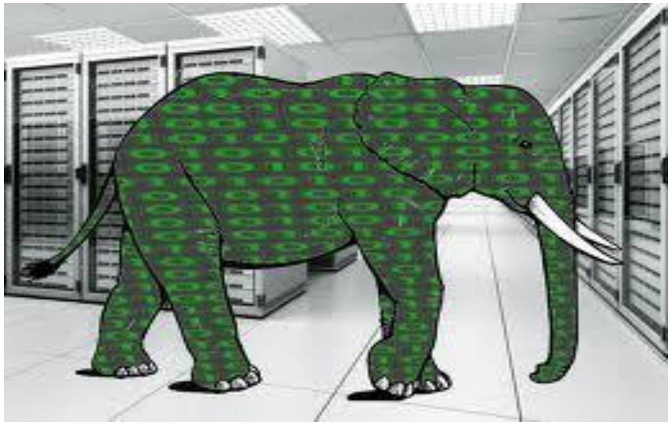
Big Data

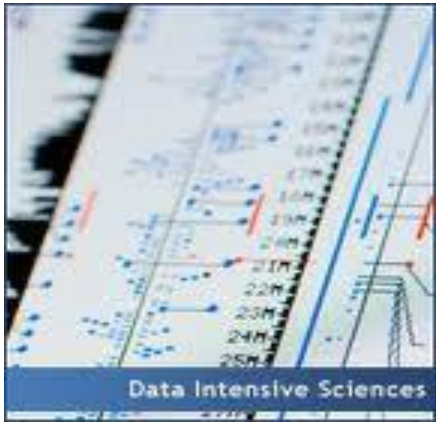
- **Big data** is defined as large pools of data that can be captured, communicated, aggregated, stored, and analyzed.

- Data continues to grow:

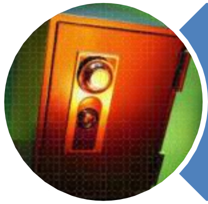
  - In mid-2010, the information universe carried 1.2 zettabytes and 2020 predictions expect nearly 44 times more at 35 zettabytes coming our way.

- Applications are becoming *data-intensive*.

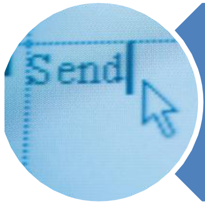




What Do We Do With Data?



Store



Share

Access

Process

Encrypt

…. and more!

We want to do these seamlessly...

Using Diverse Interfaces & Devices

Desktops

Mobile Devices

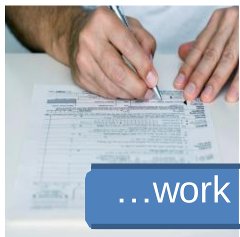
Consumer Electronics

…and even appliances

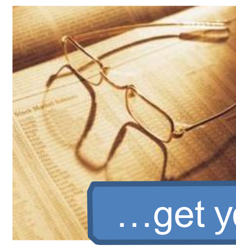
We also want to access, share and process our data from all of our devices, **anytime, anywhere**!

What About the Future?

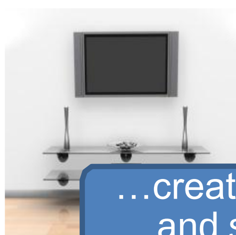
How will you…



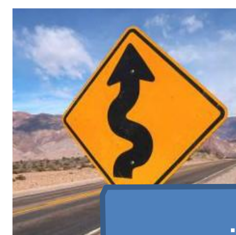
…work on documents?



…get your news & info?



…create, access, store and share media?



…navigate?

…communicate with friends and family?

…live in an intelligent home?

…

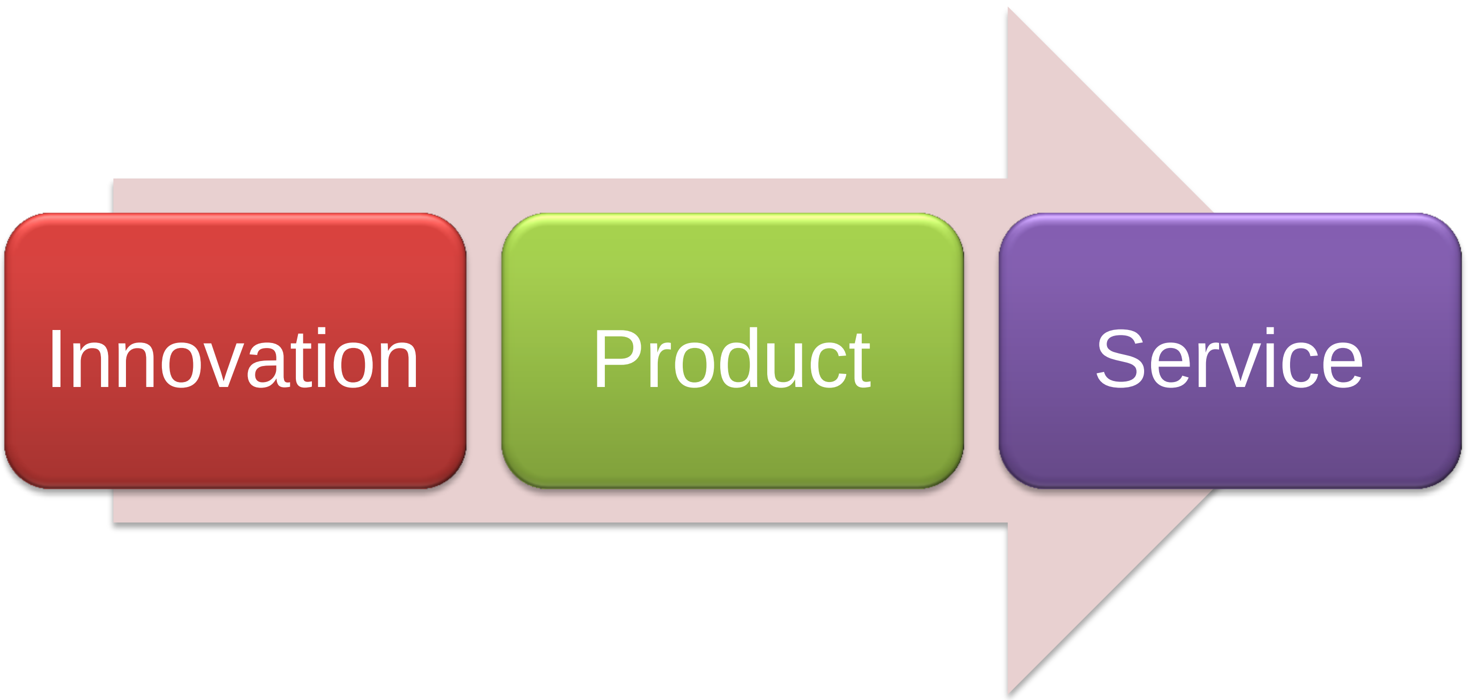

BLOOD TYPE
O-
EARTH
TUE AUG 30
10:07 AM





Obama Cabinet nominee pulls out
Israelis split Gaza in two amid calls for a cease-fire

Has this Happened Before?

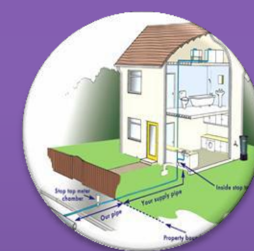


Think of it this Way …

- Evolution of water Utility

Generate your own utility

Buy it as a product



Get a continuous supply of the utility through a dedicated connection

…and Banking?

- Evolution of Banking

No Banks

(Take care of your own money)

Traditional Banking

(Give your money to the bank)

Banking Instruments

(Cheques / Credit Cards)

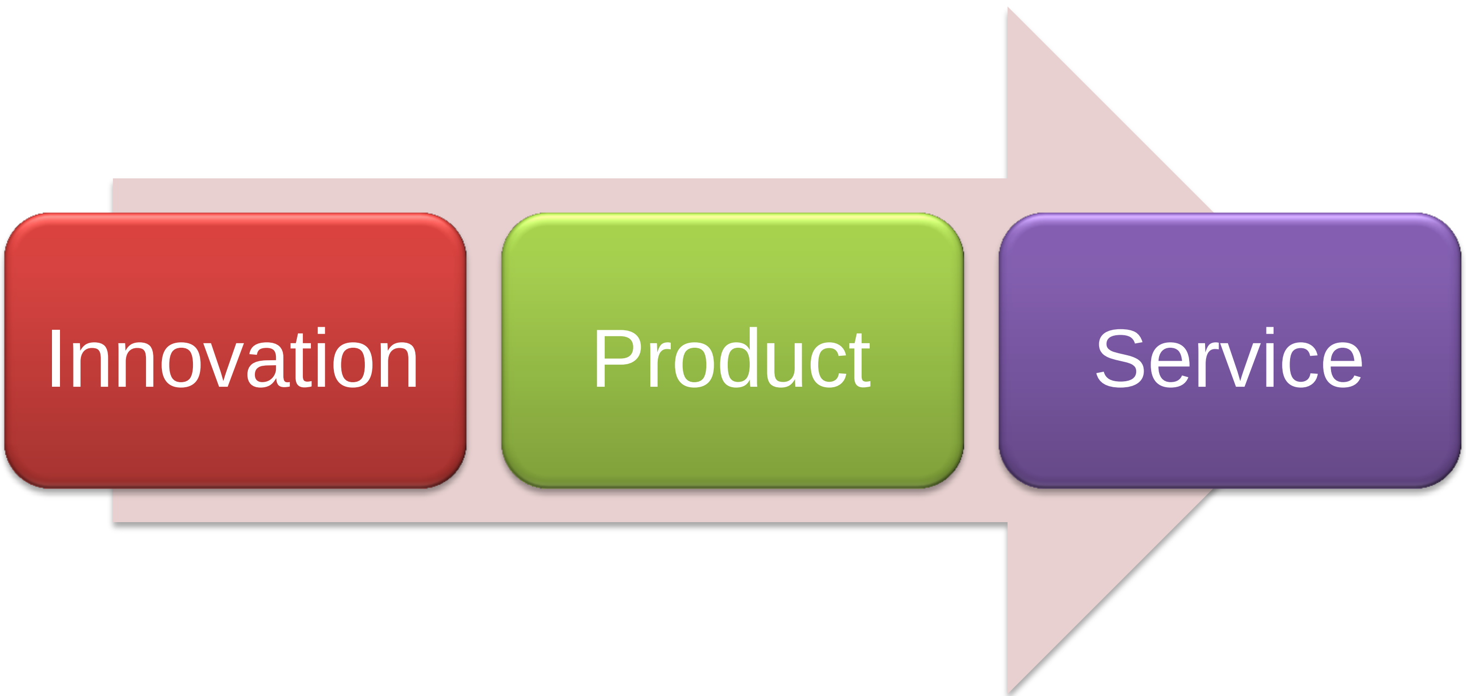
Internet Banking

(…more services)

So What is Cloud Computing?

Can We Define Cloud Computing?

"Cloud Computing is the transformation of IT from a product to a service"



Cloud Computing

- Transformation of IT from a Product to a Service

Innovation of IT

New Disruptive Technology

IT Products

Buy and Maintain the Technology

Cloud Computing

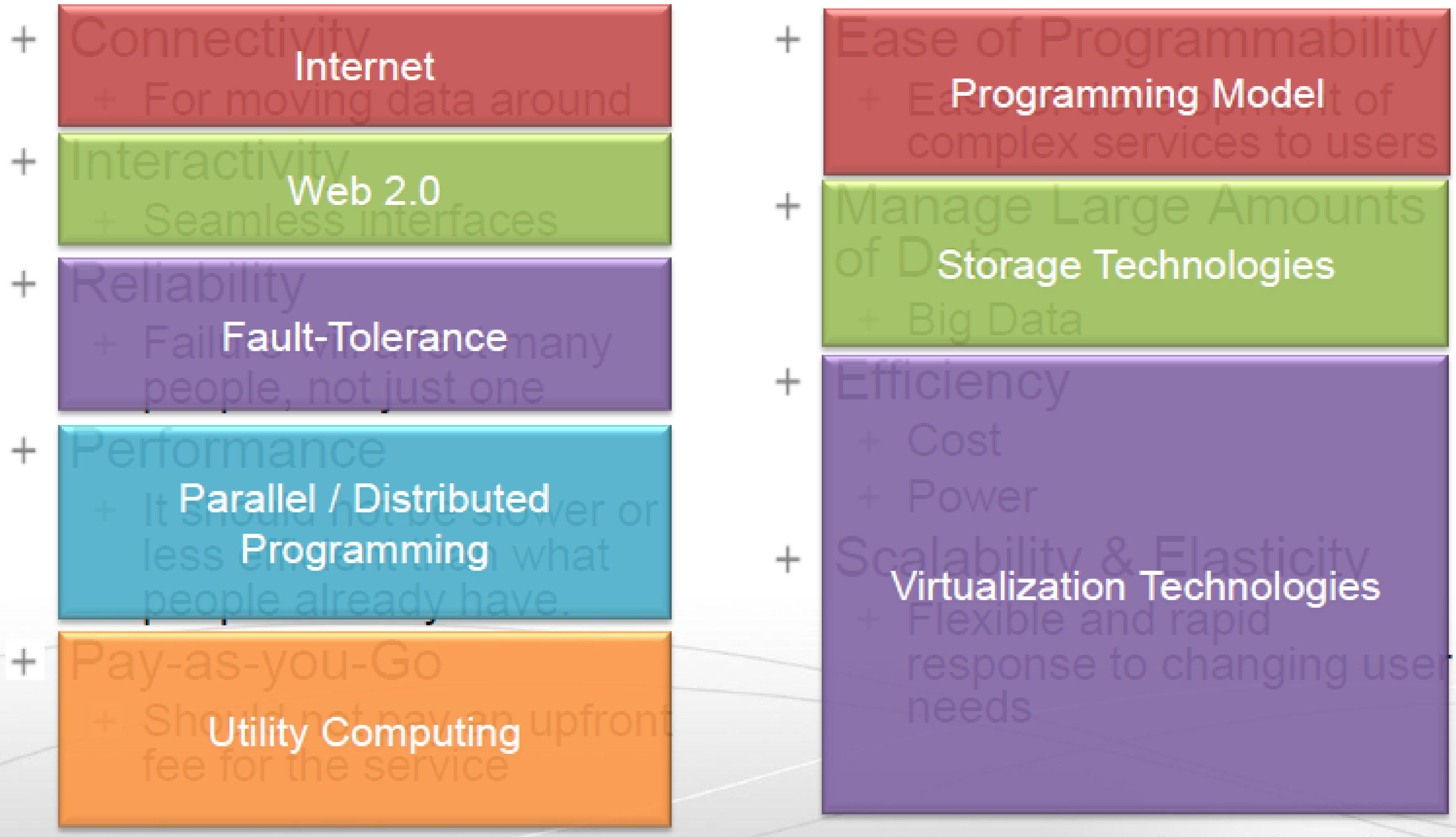
On-Demand IT services on a Pay-as You-Go basis

So… how would you transform information technology into a Service?

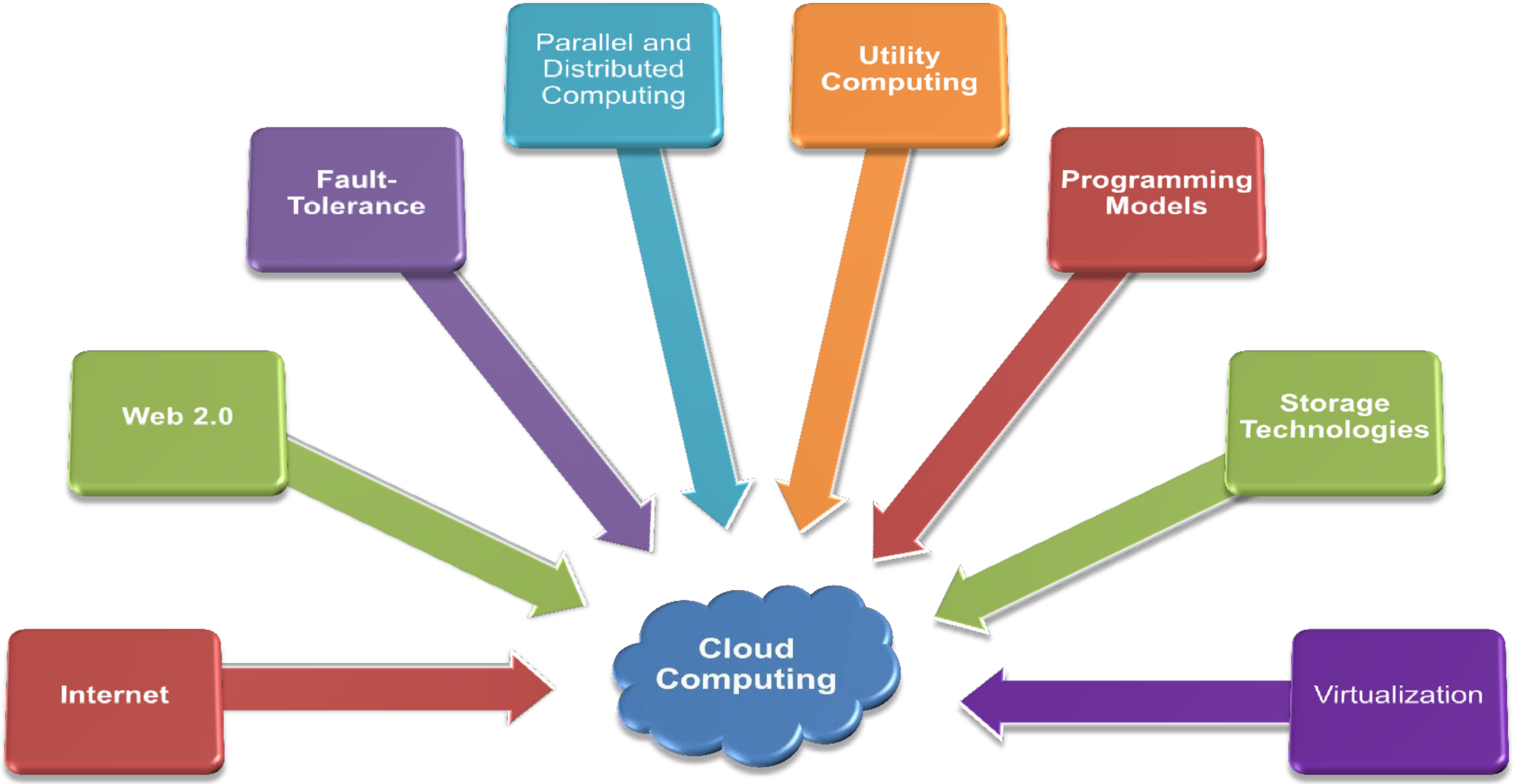
Requirements to Transform IT to a Service

- Connectivity
  - For moving data around
- Interactivity
  - Seamless interfaces
- Reliability
  - Failure will affect many people, not just one
- Performance
  - It should not be slower or less efficient than what people already have
- Pay-as-you-Go
  - Should not pay an upfront fee for the service
- Ease of Programmability
  - Ease of development of complex services to users
- Manage Large Amounts of Data
  - Big Data
- Efficiency
  - Cost Power
- Scalability & Elasticity
  - Flexible and rapid response to changing user needs

Requirements to Transform IT to a Service

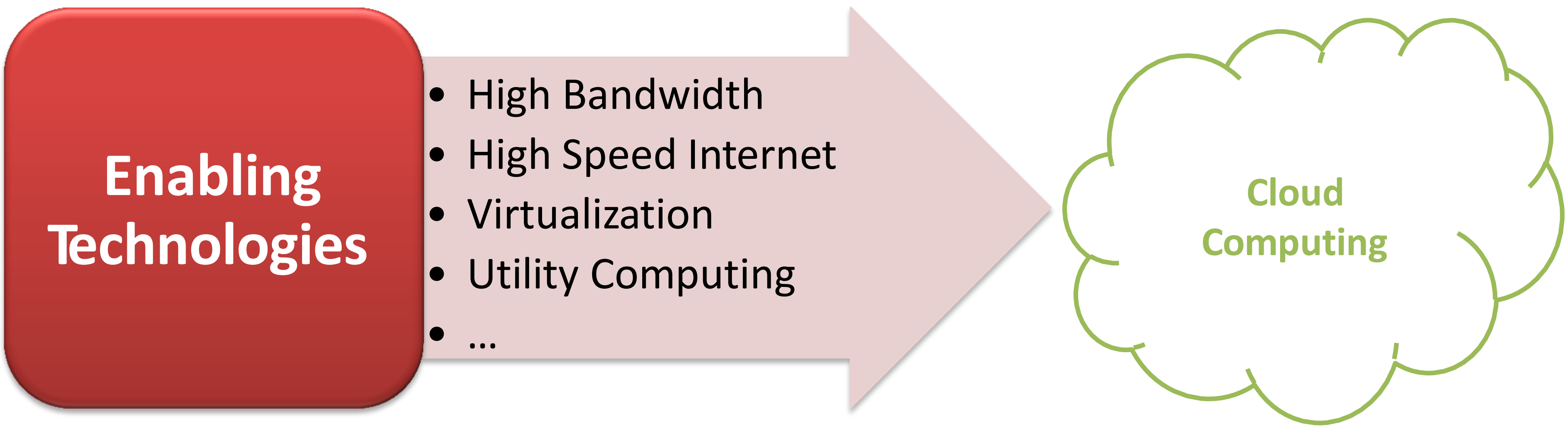


Combine the Enabling Technologies…



Cloud Computing

- Think of it as Internet Computing
  - Computation done over the Internet



… for a more complete definition!

Cloud Computing is the delivery of computing as a **service** rather than a **product**,

whereby **shared resources, software**, and **information** are provided to computers and other devices,

as a **metered service** over a **network.**

Why Cloud Computing?

Pay-as-You-Go economic model

- Reduce capital expenditure
- No upfront cost
- Reduced Time to Market

Simplified IT management

- All you need is access to the internet.
- It's the providers responsibility to manage the details.

Scale quickly and effortlessly

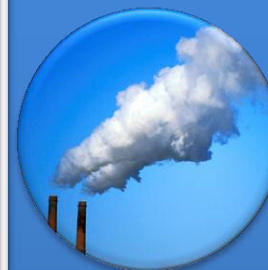
- Resources can be rented and released as required
- Software Controlled
- Instant scalablility

Flexible options

- Configure software packages, instance types operating systems.
- Any software platform
- Access from any machine connected to the Internet

Resource Utilization is improved

- Reduce Idle resources by sharing and cosolidation
- Better utilization of CPU / Storage and Bandwidth.



Carbon Footprint decreased

- Sharing of resources means less servers, less power and less emissions.

Applications Enabled by Cloud Computing

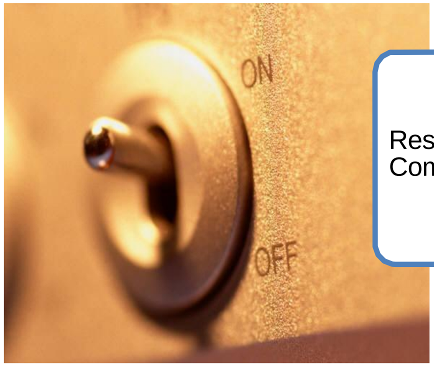
High Growth Applications

Startup Businesses

Aperiodic Bursting Applications

Seasonal Businesses

On-Off Applications



Research Computing

Periodic Applications

Changing computational patterns over time

High Growth Applications

2001

friendster

Could not keep up with the growth of their number of users.

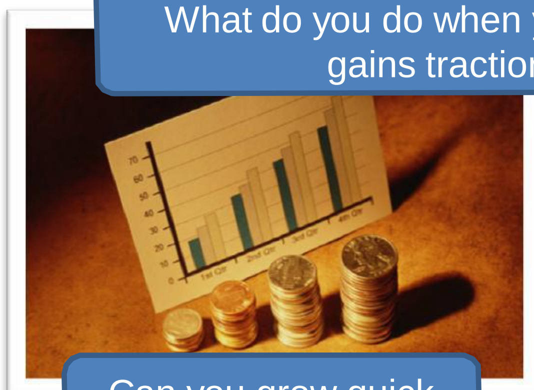
vs.

2006

facebook

Growing exponentially

What do you do when your startup gains traction?



Can you grow quick enough?

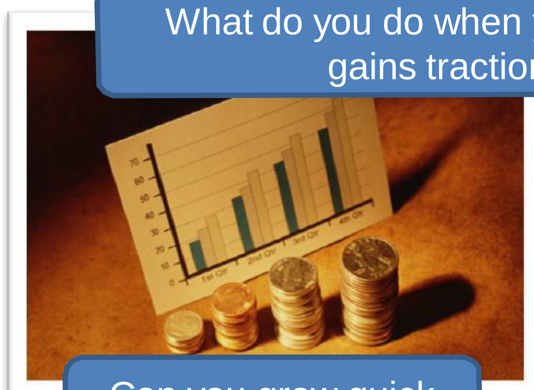
High Growth Applications

Users use it to produce video pieces from their photos, video clips and music.

Animoto's Facebook Plugin doubled traffic to the site every 12 hours for 3 days.

They could scale from 50 servers to 3500 and **go back down using cloud computing services**

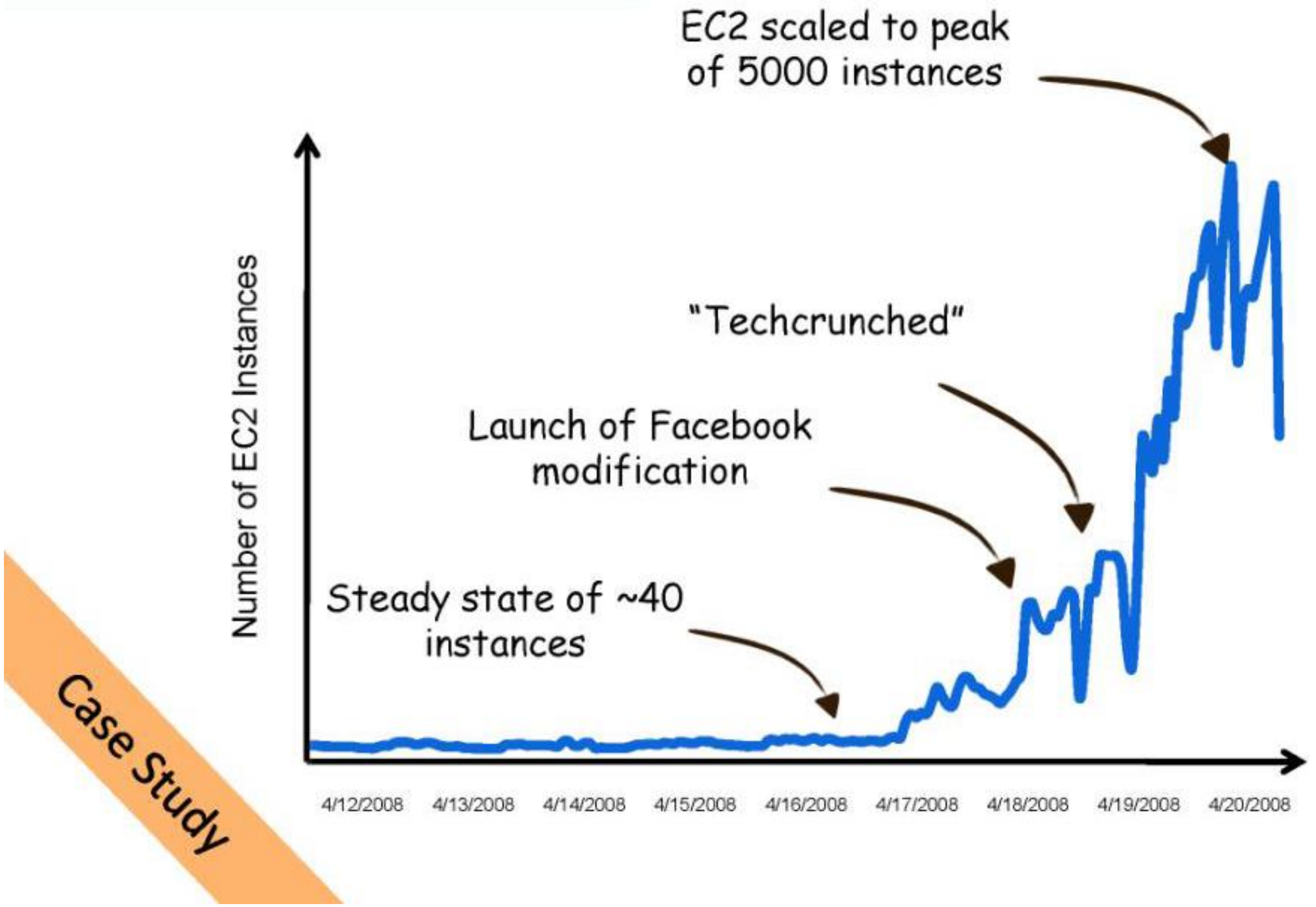
What do you do when your startup gains traction?



Can you grow quick enough?

Animoto

40 servers to 5000 in 3 days



Aperiodic Bursting Applications

CNN — Website went down on 9/11/2001 due to traffic

Hallmark — February 14th – Busiest Day of the Year

amazon.com — US Holiday Season

DOCKERS — Website crashed within 10 minutes of the free trouser promotion during Superbowl 2010

Even if you design your website infrastructure to handle peak loads, wont it be idle during other times?

On-Off Applications

- Researchers running large-scale scientific simulation using 1000s of computers.

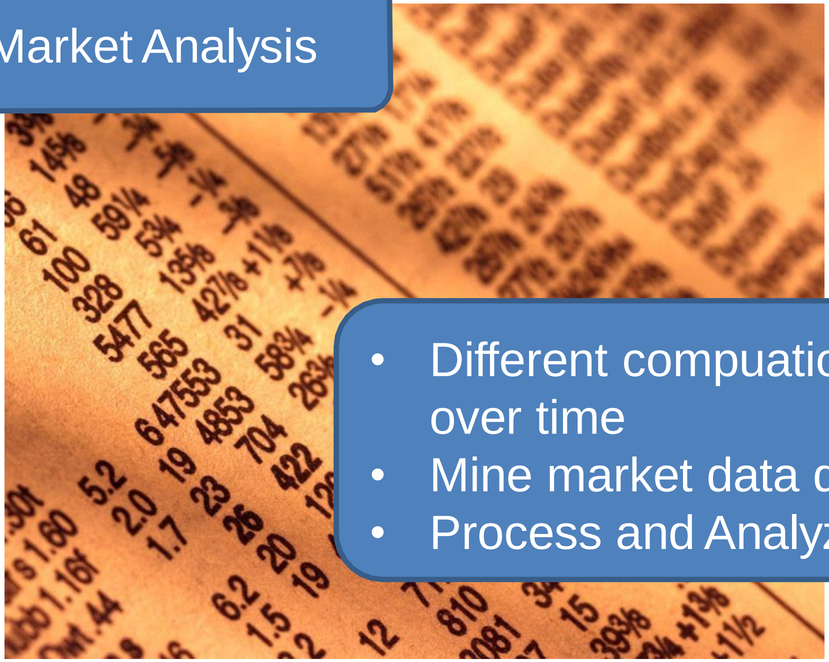
Modern Drug Discovery

- Data-intensive simulation and tests to discover new compounds
- Large compute power required for simulation jobs
- Time to market is crucial

Why not rent computer time to run these simulations?

Periodic Applications

Sock Market Analysis



- Different compuational requirements over time
- Mine market data during the day.
- Process and Analyze at night.

Dynamic and Flexible infrastructure can reduce costs and improve performance.

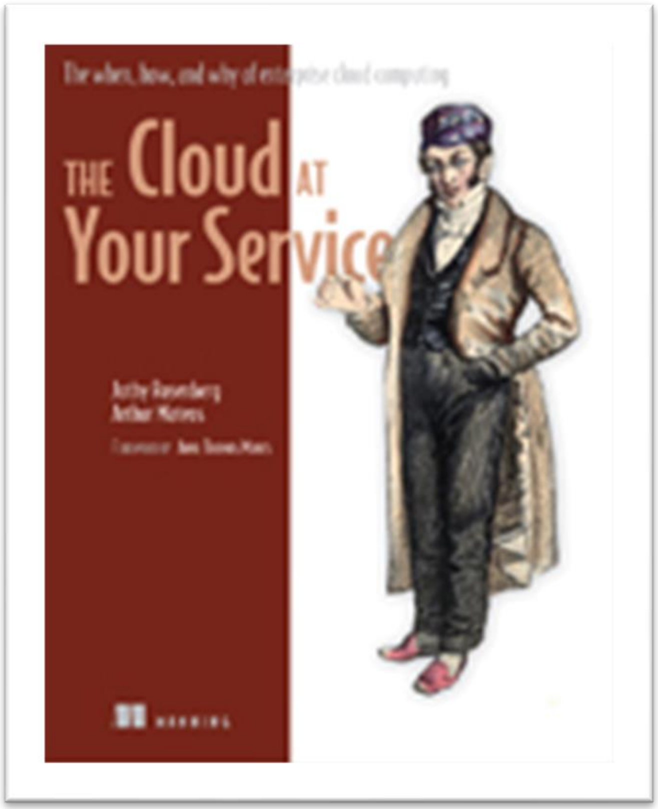
Technical Challenges

- Moving large data is still expensive
- Security
- Quality of Service
- Green computing
- Internet Dependence

Non-Technical Challenges

- Vendor Lock-In
- Non-standardized
- Security Risks
- Privacy
- Legal
- Service Level Agreements

Primary Textbook



The Cloud At Your Service

Jothy Rosenberg, Arthur Mateos
2011, Manning Publications

Reference Books

- **Virtual Machines : *Versatile Platforms for Systems and Processes***
  James E. Smith and Ravi Nair, Morgan Kauffman, 2005
- **Programming Amazon EC2**
  Jurg van Vilet and Flavia Paganelli, O'Reilly Media, 2011
- **Mahout in Action**
  Sean Owen, Robin Anil, Ted Dunning and Ellen Friedman, Manning Publications, 2011
- **Hadoop in Action**
  Chuck Lam, Manning Publications, 2011
- **More ...**

Special Note on Amazon EC2

- Paid Cloud Service – you are billed by the hour.
- Start a resource only when you need them.
- Terminate resources as soon as you are done with them.

Questions?